Teacher answer key: Primary 1

This key lists the answers and the result of the check on every page. You do not need it for routine marking: a pupil who has a working check has the right answers. To spot-check, compare the check strip with the line "Check result" below.

How the topic tags work

Each entry names the Primary 1 topic the puzzle practises, with a reference to the Primary 1 topics in the Mathematics Syllabus, Primary (2021 edition, pages 31 and 32). Where the syllabus lists several outcomes under one topic, the range of outcome numbers is given. The references are for your planning only and are not printed on the pupil pages.

Topic coverage

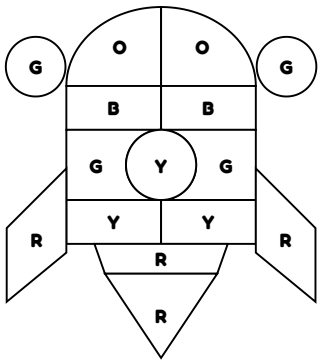
Strand	Pages
Number and Algebra - Whole Numbers	P1-01, P1-02, P1-04, P1-07, P1-14
Number and Algebra - Multiplication and Division	P1-03, P1-05, P1-13
Measurement and Geometry - Measurement	P1-06, P1-17, P1-18
Number and Algebra - Addition and Subtraction	P1-08, P1-09, P1-10, P1-11
Measurement and Geometry - Geometry	P1-12, P1-19
Number and Algebra - Money	P1-15, P1-16
Statistics - Data Analysis	P1-20

Syllabus items not covered by a puzzle: reading and writing numbers in words, number-sentence signs in isolation, measuring and drawing a line to the nearest cm, am and pm, copying figures on a grid, forming figures from shapes, ordinals beyond 5th, naming a shape from a description.

P1-01 Colour the Rocket 1 star

Topic: Place value: tens and ones

Syllabus topic: Number and Algebra - Whole Numbers; reference 1.1, 1.2 (count a set, tens and ones)
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and the parts in the same place on the left and on the right match.
Answers: Colour key: 1 = yellow, 2 = red, 3 = blue, 4 = green, 5 = orange
Check result: Tally: yellow 3, red 4, blue 2, green 4, orange 2; the parts in the same place on the left and on the right match.



Teacher note: Every part asks a tens-and-ones question: how many ones or tens a number has, or the missing part of a number made of tens and ones. The answers are 1 to 5. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); the parts in the same place on the left and on the right match.

P1-02 Mystery Numbers 2 stars

Topic: Numbers to 100: tens digit, ones digit, more and less

Syllabus topic: Number and Algebra - Whole Numbers; reference 1.2, 1.5 (tens and ones, compare and order numbers)
Puzzle type: Number riddles. How the pupil checks: Code word (5 letters) and testing each clue.
Answers: 1) 66, 2) 85, 3) 36, 4) 56, 5) 26
Check result: Code word: JUICE
Teacher note: Every riddle has 3 clues and every clue is needed. A pupil who stops after one clue gets more than one possible number; testing the other clues removes them. Clues used on this page: more than, less than, ones digit, tens digit.

P1-03 Fair Shares 1 star

Topic: Sharing equally and making equal groups (within 20)

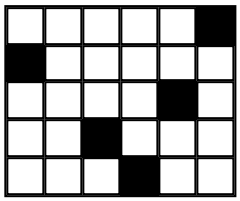
Syllabus topic: Number and Algebra - Multiplication and Division; reference 3.4 (dividing within 20: sharing and grouping, no division sign)
Puzzle type: Code-breaker. How the pupil checks: Code word (6 letters) and multiplying back.
Answers: 1) 6, 2) 4, 3) 3, 4) 2, 5) 8, 6) 5
Check result: Code word: SHADOW
Teacher note: Pictures of apples, stars, fish and balloons, with empty plates, boxes, bowls and bags. Three items share the objects equally among the containers drawn; three put the objects into groups of a stated size (a dashed loop shows the first group). Every share is exact, with nothing left over. No division sign is used. A pupil can check by multiplying back: containers times each one's share is the number of objects.

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P1-04 Sticker Grid 2 stars

Topic: Comparing numbers to 100 (more, fewer, greatest, smallest)

Syllabus topic: Number and Algebra - Whole Numbers; reference 1.5 (compare and order numbers)
Puzzle type: Logic grid. How the pupil checks: Letters beside the numbers, read down the list, spell a word (5 letters; one of the six numbers is not used).
Answers: Ali: 99 (F), Mei: 12 (R), Siti: 88 (U), Ben: 63 (I), Kumar: 78 (T); nobody has 23 (letter P is the decoy)
Check result: Letters down the list: FRUIT



Teacher note: Six numbers are listed but there are only five children: a clue says which number nobody has, and its letter (P) is a decoy that is not part of the word. The five letters of the word have exactly one arrangement that is a word, so no wrong placement of the numbers spells a word: all 719 wrong ways to give the children five of the six numbers were tested (checked against a list of about 129 000 words). A pupil whose letters are not a word must go back to the clues. The page shows a worked example of crosses and ticks. Clues on this page: more than, fewer than, greatest, second greatest.

P1-05 Times Code 2 stars

Topic: Multiplication within 40: groups of, and the missing number

Syllabus topic: Number and Algebra - Multiplication and Division; reference 3.1 to 3.3 (concept of multiplication, sign x, multiplying within 40)
Puzzle type: Code-breaker. How the pupil checks: Code word (6 letters) and putting the number back.
Answers: 1) 20, 2) 4, 3) 5, 4) 30, 5) 6, 6) 10
Check result: Code word: SQUARE
Teacher note: Second multiplication page (the equal groups page uses pictures). Two items give the product with the sign, two use the words groups of, and two have a missing number in a multiplication sentence, which is the two-star step: the pupil must undo the sum. Factors are 2, 3, 4, 5 and 10 and every product is within 40.

P1-06 What Time Is It? 1 star

Topic: Time: o'clock and half past

Syllabus topic: Measurement and Geometry - Measurement; reference Time 2.1 (tell time: o'clock and half past)
Puzzle type: Code-breaker. How the pupil checks: Code word (6 letters).
Answers: 1) 8:00 (8 o'clock), 2) 4:00 (4 o'clock), 3) 9:00 (9 o'clock), 4) 1:30 (half past 1), 5) 11:30 (half past 11), 6) 2:30 (half past 2)
Check result: Code word: SECOND
Teacher note: Three o'clock times and three half past times; the short hand is never close to the long hand. The code table holds the six right times and four wrong readings (the hour one too late, the hour one too early, o'clock read as half past, or the other way round). Times to 5 minutes are on the Clock Time page.

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P1-07 The Queue 2 stars

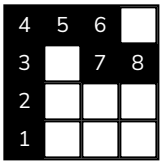
Topic: Ordinal numbers: 1st to 5th, next to

Syllabus topic: Number and Algebra - Whole Numbers; reference 1.7 (ordinal numbers first to tenth)
Puzzle type: Logic (queue). How the pupil checks: Shirt letters read from 1st to 5th spell a word (5 letters; one child is not in the queue).
Answers: 1st Jenny, 2nd Aisha, 3rd Ali, 4th Ravi, 5th Lily; Kumar is not in the queue (letter W is the decoy)
Check result: Word: ENJOY
Teacher note: Six children are shown but the queue has five places: a clue says who is not in the queue, and that child's shirt letter (W) is a decoy that is not part of the word. The top row is not in queue order. The five letters of the word have exactly one arrangement that is a word, so no wrong queue spells a word: all 719 wrong queues of five of the six children were tested (checked against a list of about 129 000 words). Clues on this page: ordinal numbers (3rd, 5th), next to. Two or more clues compare two children, so the pupil must combine them. Syllabus ordinals go to 10th; this page uses 1st to 5th.

P1-08 Make Twelve 1 star

Topic: Addition within 20

Syllabus topic: Number and Algebra - Addition and Subtraction; reference 2.1 to 2.7 (add within 20, mental calculation)
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box making 12.
Answers: Route (row, column from top-left): (4,1) 8+4 -> (3,1) 5+7 -> (2,1) 9+3 -> (1,1) 6+6 -> (1,2) 6+6 -> (1,3) 5+7 -> (2,3) 9+3 -> (2,4) 4+8
Check result: Route reaches FINISH through 8 boxes; every box on it adds to 12.



Teacher note: A 4 by 4 grid: sixteen sums, all within 20. Boxes off the route make 8, 10, 11, 14 and a slip in adding sends the pupil there. 2 extra boxes that do make 12 are dead ends. Counting the boxes shows a line that takes a shortcut.

P1-09 Magic Square 2 stars

Topic: Adding three 1-digit numbers

Syllabus topic: Number and Algebra - Addition and Subtraction; reference 2.1 to 2.7 (adding more than two 1-digit numbers)
Puzzle type: Magic square. How the pupil checks: All eight line totals make 15.
Answers: Rows: 4 9 2 / 3 5 7 / 8 1 6
Check result: All 8 totals = 15

4	**9**	**2**
3	**5**	**7**
8	**1**	**6**

Teacher note: Only three numbers are given. The solver shows the other six are forced. The sum boxes let a pupil see which line is wrong. Each number 1 to 9 must be used once.

P1-10 Take Away Code 1 star

Topic: Subtraction within 20; missing numbers

Syllabus topic: Number and Algebra - Addition and Subtraction; reference 2.1 to 2.7 (subtract within 20, inverse relationship)
Puzzle type: Code-breaker. How the pupil checks: Code word (6 letters) and putting the number back.
Answers: 1) 9, 2) 7, 3) 4, 4) 8, 5) 5, 6) 16
Check result: Code word: COUSIN
Teacher note: Items 5 and 6 ask for the number taken away. Pupils can check by adding back, which is the inverse idea.

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P1-11 Big Number Sums 2 stars

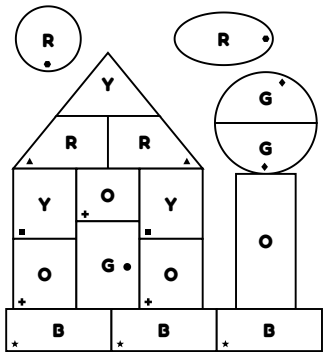
Topic: Add and subtract within 100

Syllabus topic: Number and Algebra - Addition and Subtraction; reference 2.1 to 2.7 (add and subtract within 100: 2-digit and ones, 2-digit and tens, missing number)
Puzzle type: Code-breaker. How the pupil checks: Code word (6 letters) and putting the number back.
Answers: 1) 21, 2) 81, 3) 94, 4) 30, 5) 17, 6) 88
Check result: Code word: FAMILY
Teacher note: Numbers are within 100 and no sum crosses a ten (no renaming). Items: 2-digit plus ones, 2-digit plus tens, take away ones, take away tens, and two missing-number items. The missing-number items are the two-star step: the pupil must undo the sum.

P1-12 Colour the House 1 star

Topic: Shapes: corners

Syllabus topic: Measurement and Geometry - Geometry; reference Geometry 1.1 (identify, name, describe 2D shapes: triangle, square, rectangle, circle)
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and parts with the same small mark match.
Answers: Colour key: 3 = yellow, 4 = red, 6 = blue, 7 = green, 8 = orange
Check result: Tally: yellow 3, red 4, blue 3, green 3, orange 4; parts with the same small mark match.



Teacher note: Shapes are triangle, square, rectangle and circle only. A circle has no corners. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); parts with the same small mark match.

P1-13 Equal Groups 1 star

Topic: Multiplication: equal groups

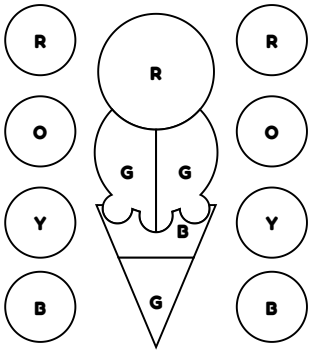
Syllabus topic: Number and Algebra - Multiplication and Division; reference 3.1 to 3.3 (concept of multiplication, sign x, multiplying within 40)
Puzzle type: Code-breaker. How the pupil checks: Code word (6 letters).
Answers: 1) 2 bags of 5 = 10, 2) 5 bags of 5 = 25, 3) 3 bags of 6 = 18, 4) 3 bags of 8 = 24, 5) 2 bags of 8 = 16, 6) 5 bags of 4 = 20
Check result: Code word: NUMBER
Teacher note: Pictures of equal groups. Pupils write the number of groups, the number in each group and the total. Totals are within 40.

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P1-14 Colour the Ice Cream 2 stars

Topic: Counting in 2s, 5s and 10s

Syllabus topic: Number and Algebra - Whole Numbers; reference 1.6 (number patterns); counting in 2s, 5s and 10s leads to multiplication
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and the parts in the same place on the left and on the right match.
Answers: Colour key: 60 = yellow, 65 = red, 70 = blue, 75 = green, 80 = orange
Check result: Tally: yellow 2, red 3, blue 3, green 3, orange 2; the parts in the same place on the left and on the right match.



Teacher note: Patterns count in 2s, 5s or 10s, forwards or back, and the answers are 60 to 80. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); the parts in the same place on the left and on the right match.

P1-15 Coin Purses 1 star

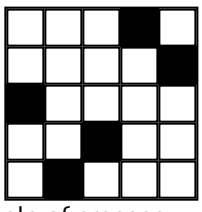
Topic: Money: counting coins in cents (up to $1)

Syllabus topic: Number and Algebra - Money; reference Money (count money in cents up to $1)
Puzzle type: Code-breaker. How the pupil checks: Code word (5 letters).
Answers: 1) 80¢, 2) 60¢, 3) 40¢, 4) 25¢, 5) 95¢
Check result: Code word: PIANO
Teacher note: Coins used: 5¢, 10¢, 20¢, 50¢. Each coin shows its value and the word cents. The coins are original line drawings (a small picture on each, not a copy of any real coin), with sizes in the true proportion of the real coins. Totals stay below $1.

P1-16 Shop Prices 2 stars

Topic: Money: dollars, more and less

Syllabus topic: Number and Algebra - Money; reference Money (dollars up to $100) with 2.1 to 2.7 (add, subtract)
Puzzle type: Logic grid + total. How the pupil checks: Letters beside the prices, then a code letter for the total, spell a word (6 letters).
Answers: pen: $12 (F), book: $15 (R), ball: $5 (O), kite: $8 (Z), cup: $7 (E); the ball and the pen cost $17 together (code letter N)
Check result: Letters in order: FROZEN



Teacher note: Five items, five prices, each price with a letter. The page shows a worked example of crosses and ticks. '$3 more than' means a price $3 higher: pupils test each price against it. The last answer is added from the grid and looked up in a small code table, so a wrong grid or a wrong sum changes the word.

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P1-17 How Long? 1 star

Topic: Length: measure in centimetres

Syllabus topic: Measurement and Geometry - Measurement; reference Length 1.1, 1.2 (measure in cm, abbreviation cm)
Puzzle type: Code-breaker. How the pupil checks: Code word (6 letters).
Answers: 1) 4 cm, 2) 5 cm, 3) 3 cm, 4) 2 cm, 5) 7 cm, 6) 8 cm
Check result: Code word: LENGTH
Teacher note: A ruler is drawn under each strip at 10 mm to 1 cm. If the page is copied at a different scale the strip and ruler change together, so the reading from the drawn ruler stays right, but a real ruler laid on the page will not match: print at 100% (actual size) if pupils will use a real ruler. Two strips start at 0 and four start at 1 to 4 on the ruler; a bold arrow and a bold tick show the start. The Primary 1 length outcomes list measuring in cm and do not list measuring from a mark other than 0, so the offset strips are a small extension: the pupil counts the cm from the arrow to the end of the strip. Every length is a whole number of cm.

P1-18 Clock Time 2 stars

Topic: Time to 5 minutes: 1 hour and half an hour later

Syllabus topic: Measurement and Geometry - Measurement; reference Time 2.1 and 2.4 (tell time to 5 minutes; durations of 1 hour or half an hour)
Puzzle type: Code-breaker. How the pupil checks: Code word (5 letters).
Answers: 1) shows 8:10, 1 hour later 9:10, 2) shows 1:25, half an hour later 1:55, 3) shows 1:30 (half past 1), 1 hour later 2:30 (half past 2), 4) shows 11:25, half an hour later 11:55, 5) shows 11:00 (11 o'clock), 1 hour later 12:00 (12 o'clock)
Check result: Code word: CLIMB
Teacher note: Times to 5 minutes are usually taught late in Primary 1 (use the What Time Is It? page for o'clock and half past). Each item is two steps: read the clock, then add 1 hour or half an hour. None of the half-hour additions cross the hour. The decoy times in the table are the hour one too late, the hands read the wrong way round. The hands are never closer than 30 degrees. a.m. and p.m. are not used on this page.

P1-19 Shape Count 1 star

Topic: 2D shapes: rectangle, triangle, circle, half circle, quarter circle

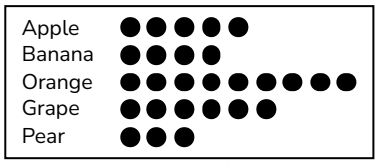
Syllabus topic: Measurement and Geometry - Geometry; reference Geometry 1.1, 1.3 (name 2D shapes; identify the shapes in a figure)
Puzzle type: Count + code-breaker. How the pupil checks: The five counts add to 15, and a code word (5 letters).
Answers: rectangles: 2, triangles: 4, circles: 5, half circles: 3, quarter circles: 1 (total 15)
Check result: Total 15; code word: BUILD
Teacher note: The picture has no squares so there is no 'is a square a rectangle?' question. Shapes are turned in different directions on purpose. The five counts are all different (1 to 5) so each count gives one letter.

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P1-20 Fruit Graph 2 stars

Topic: Picture graphs: read, complete and answer

Syllabus topic: Statistics - Data Analysis; reference Picture graphs (read and interpret picture graphs)
Puzzle type: Graph detective. How the pupil checks: Two sums: the total of all the faces, and the row chosen in a) plus the answer to b) (neither is used to find any row).
Answers: Apple: 5, Banana: 4, Orange: 9, Grape: 6, Pear: 3; a) Orange; b) 2
Check result: Total 27 faces; second sum 9 + 2 = 11

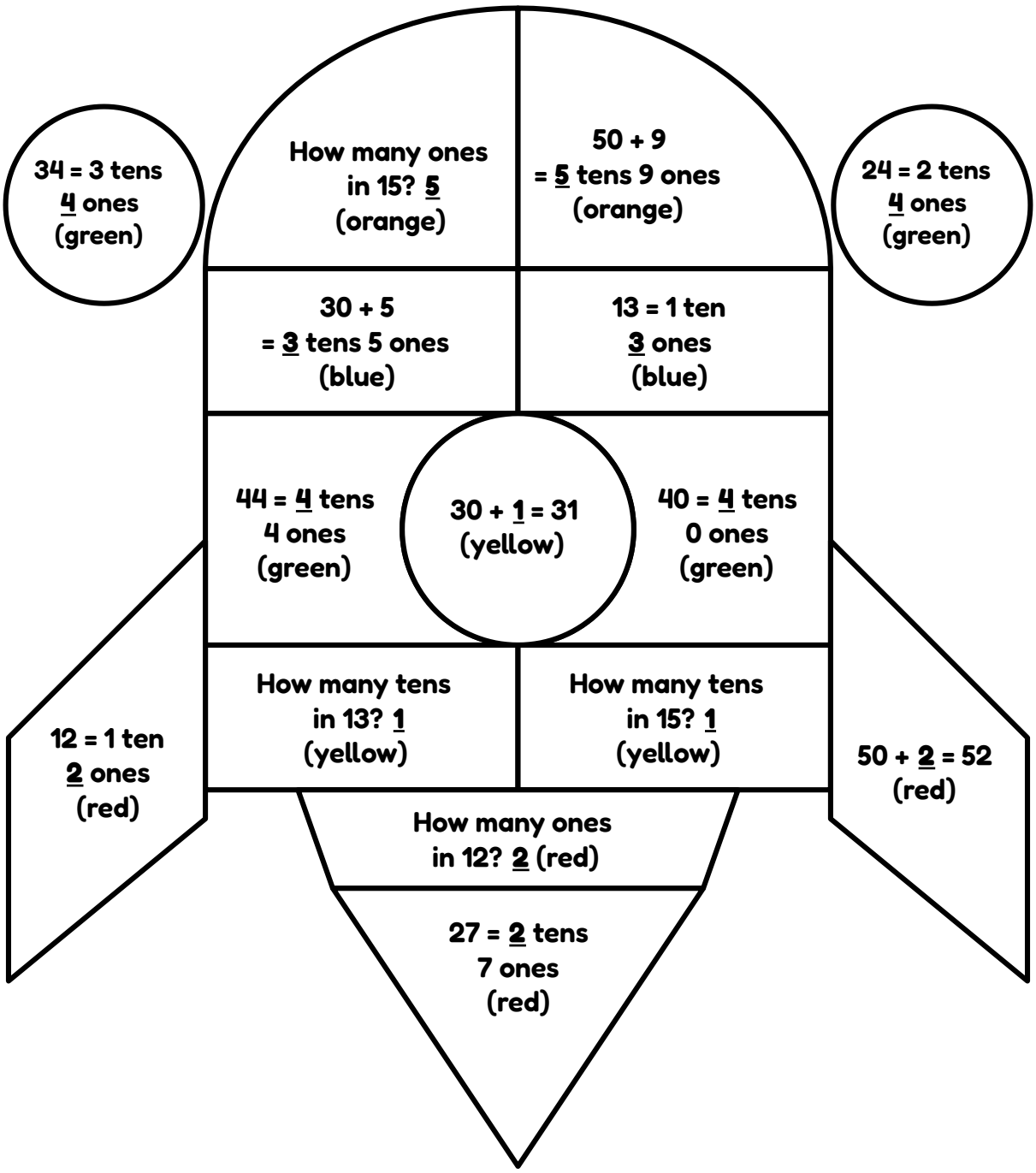


Teacher note: Each smiley face stands for one child (a scale of 1: one picture symbol for one child). A pupil who draws a plain circle instead of a face counts the same, so the checks do not change. The three clues fix the three missing rows one by one. The total of 27 is printed only in the check strip, so it is a separate test: a pupil who misreads a clue or draws one circle too many or too few gets a different total. The second sum tests a) and b): it is 11 only when the right row is chosen and b) is right. The three clues start from Apple or Banana; a pupil who starts a clue from the wrong one of the two, or who reads more as fewer, also gets a wrong total or a wrong second sum (every such mix was tried). Question b) names two other fruits, so it does not give away the answer to a). More and fewer are not mirror images here (the Grape clue is 2 and the Pear clue is 1), and the first two clues add different amounts (4 and 2), so a pupil who swaps the two words or swaps Apple and Banana gets a different total or a different second sum.

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Answers: P1-01 Colour the Rocket

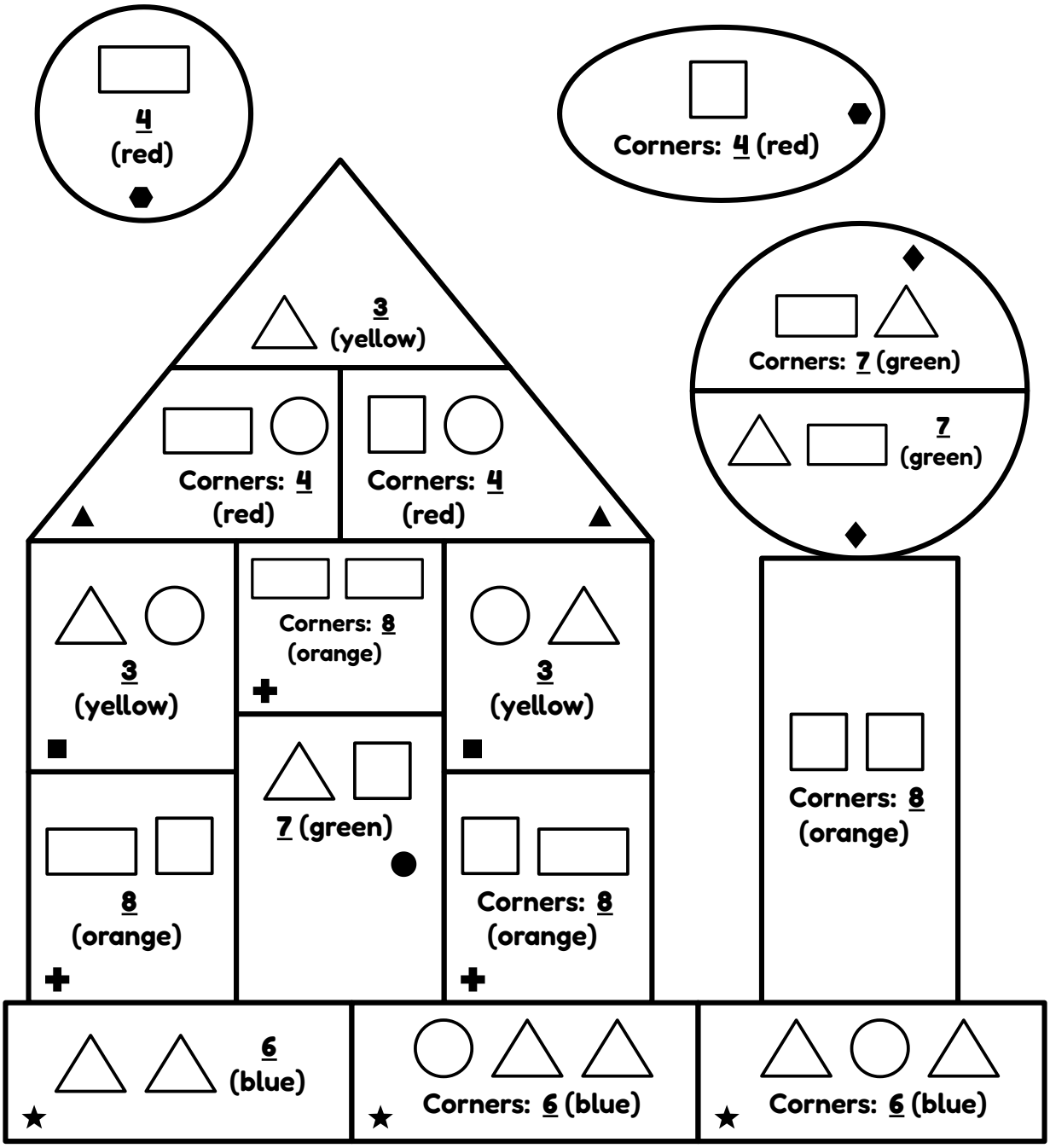
Every part shows the question, the answer underlined and the colour in brackets. Pupils colour their own copy. Colour key: 1 = yellow, 2 = red, 3 = blue, 4 = green, 5 = orange.



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Answers: P1-12 Colour the House

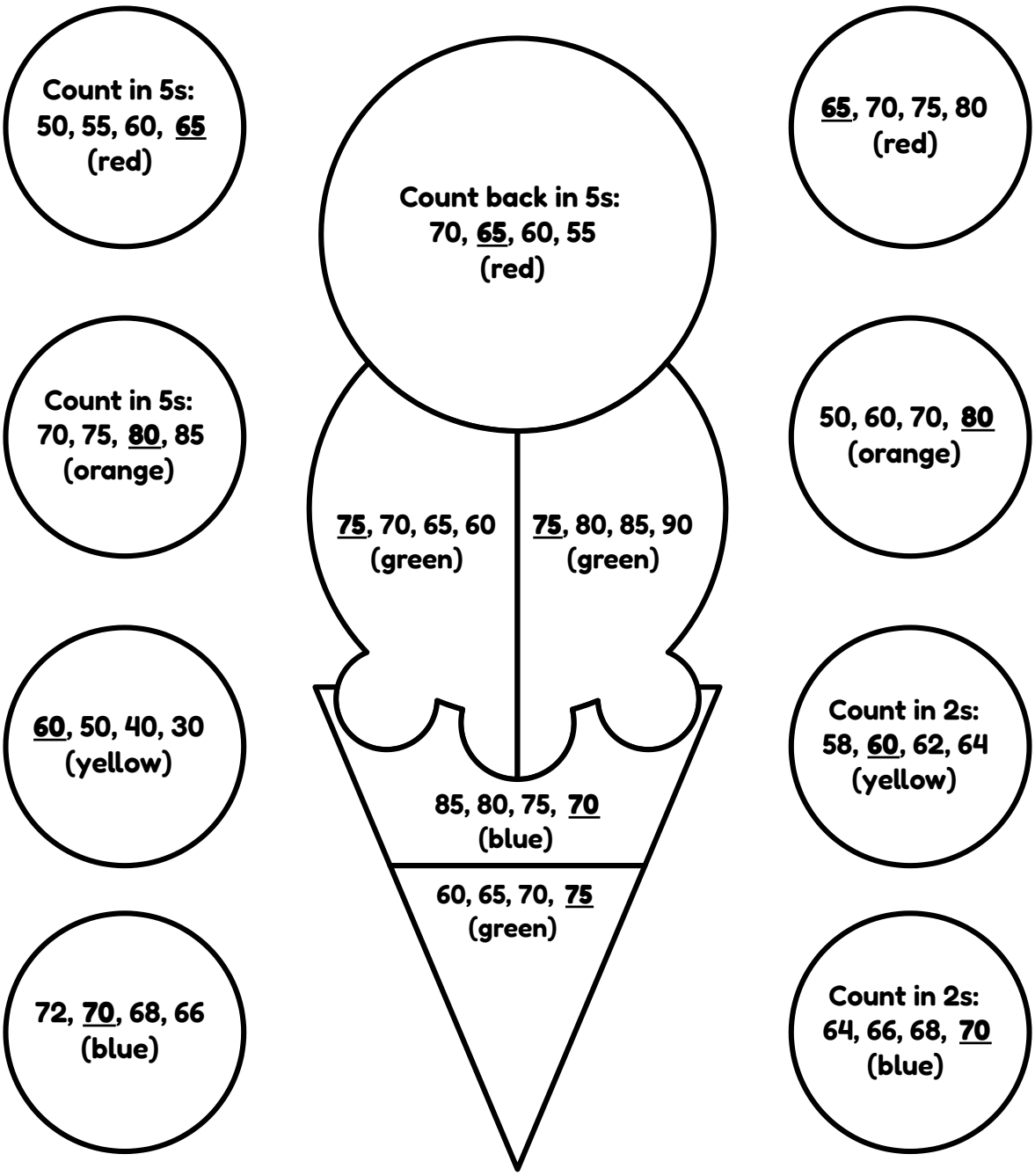
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Answers: P1-14 Colour the Ice Cream

Every part shows the question, the answer underlined and the colour in brackets. Pupils colour their own copy. Colour key: 60 = yellow, 65 = red, 70 = blue, 75 = green, 80 = orange.



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